

Chapter 7

Basic Portfolio Management

Whether adopting an equity, option, or hybrid portfolio, building a portfolio is nontrivial. Identifying a suitable collection of elements, calculating optimal portfolio weights, and maintaining that balance easily becomes hairy. Though countless ways to approach this process exist, the portfolio management tactics discussed in this book are fairly back-of-the-envelope and divided between two chapters. This chapter covers *necessary* guidelines in portfolio management, and the following chapter covers advanced portfolio management including *supplementary* techniques for portfolio optimization. Basic portfolio management includes the following concepts:

- Capital allocation guidelines
- Diversification
- Maintaining portfolio Greeks

Capital Allocation and Position Sizing

The purpose of the dynamic allocation guidelines first introduced in [Chapter 3](#) is to limit portfolio tail exposure while also allowing for reasonable long-term growth by scaling capital allocation according to the current risks and opportunities in the market. Recall that the amount of portfolio buying power allotted to short premium positions, such as short strangles and short iron condors, should range from 25% to 50%, depending on the current market volatility, with the remaining capital either kept in cash or a low-risk passive investment. Of the amount allocated to short premium, at least 75% should be reserved for undefined risk trades (with less than 7% of portfolio buying power allocated to a single position) and at most 25% reserved for defined risk strategies (with less than 5% of portfolio buying power allocated to a single position), although there are exceptions for high probability of profit (POP), defined risk strategies. It's worth mentioning that it is not always feasible to strictly abide by the position size caps of 5% to 7%. If a portfolio has only \$10,000 in buying

power and implied volatility (IV) is low (i.e., $VIX < 15$), this rule limits the maximum per-trade buying power reduction (BPR) to \$700 for an undefined risk trade at a time when BPRs tend to be high. This guideline would severely limit the opportunities available for small accounts. Though total portfolio allocation guidelines *must* be followed, there is more leniency for the per-trade allocation guidelines in smaller accounts.

These guidelines limit the amount of capital exposed to outlier losses, but how capital is allocated depends on personal profit goals and loss tolerances. An options portfolio is typically composed of two types of positions: core and supplemental. Core positions are usually high-POP trades with moderate profit and loss (P/L) standard deviation. These types of positions should offer consistent, fairly reliable profits and reasonable outlier exposure although they will vary by risk tolerance. Consider the following examples:

- Riskier core position: a 45 days to expiration (DTE) 20 Δ strangle (undefined risk trade) with a diversified exchange-traded fund (ETF) underlying, such as SPY or QQQ.
- More conservative core position: a 45 DTE 16 Δ SPY iron condor with 6 Δ wings (high-POP, defined risk trade) with a diversified ETF underlying, such as SPY or QQQ.

Core positions should comprise the majority of a portfolio and be diversified across sectors to develop more reliable portfolio profit and loss expectations and resilience to market volatility. Supplemental positions are not necessarily dependable sources of profit but rather tools for market engagement. These positions are typically higher-risk, higher-reward trades meant to capitalize on dynamic opportunities in the market. Some examples of supplemental positions include earnings trades (which will be discussed in more detail in [Chapter 9](#)) or strangles with stock underlyings, such as a 45 DTE 16 Δ AAPL strangle. When trading stock underlyings, defined risk supplemental positions would be suitable for more risk-averse traders. These types of positions have significantly more P/L variability than positions with ETF underlyings, resulting in more per-trade profit potential and more loss potential with less dependable profit and loss expectations.

The expected returns, P/L variability, and tail exposure of a portfolio overall primarily depend on the types of core positions, types of supplemental positions, and the ratio of core positions to supplemental positions.

Portfolios for more risk-tolerant traders may include a larger percentage of supplemental positions. However, mitigating tail risk remains the highest priority, particularly if the portfolio underlyings are not diversified well. This is why, generally speaking, at most 25% of the capital allocated to short premium should go toward supplemental positions. For example, if the VIX is valued at 45 and 50% of portfolio buying power is allotted to short premium positions (per the allocation guidelines), then at most 25% of the 50% portfolio buying power (or 12.5%) should be allocated to supplemental positions. See [Table 7.1](#) for some numerical context.

Compared to core positions, such as SPY or QQQ strangles, the supplemental positions above have significantly more profit potential, loss potential, and tail risk exposure. The average profit is larger partially as the result of supplemental underlying assets having higher per share prices. This was the case with GOOGL and AMZN, which cost more than the other equity underlyings throughout the entire backtest period. However, these instruments also carry larger profit potentials as option underlyings because they are subject to company-specific risk factors that often inflate the values of their respective options. This was particularly the case with AAPL, which had a *lower* per share value than SPY, QQQ, and GLD throughout this backtest period but more option volatility.

Table 7.1 Statistics for 45 DTE 16 Δ strangles from 2011–2020, managed at expiration. Included are examples for core and supplemental position underlyings.

	16 Δ Strangle Statistics (2011–2020)				
	Underlying	POP	Average Profit	Average Loss	Conditional Value at Risk (CVaR) (5%)
Core	SLV	84%	\$32	–\$88	–\$201
	QQQ	74%	\$109	–\$183	–\$454
	SPY	80%	\$162	–\$320	–\$800
	GLD	81%	\$119	–\$456	–\$1,100
Supplemental	AAPL	74%	\$425	–\$1,443	–\$4,771
	GOOGL	80%	\$1,174	–\$2,955	–\$6,593
	AMZN	77%	\$1,235	–\$2,513	–\$6,810

These statistics do not account for IV or stock-specific factors, such as earnings or dividends.

Due to these single-stock risk factors and the variance reflected in the option P/Ls, stocks are generally unsuitable underlyings for core positions. Their high profit potentials make them appealing supplemental position underlyings for opportunistic investors, but mitigating the tail risk exposure from supplemental positions is key for portfolio longevity. The most effective way to accomplish this is by strictly limiting the portfolio capital allocated to high-risk positions.

To summarize, core positions should provide somewhat reliable expectations around P/L and be diversified across sectors. Supplemental positions should comprise a smaller percent of a portfolio because they bring higher profit potentials but also more risk. Diversification, particularly when trading options, is another crucial risk management strategy that can significantly reduce portfolio P/L variability and outlier exposure.

The Basics of Diversification

All financial instruments are subject to some degree of risk, with the risk profiles of some instruments being more flexible than others. A single equity has an immutable risk profile, and an option's risk profile can be adjusted according to multiple parameters. However in either scenario, traders are subject to the risk factors of the particular position. When trading a *portfolio* of assets, a trader may offset the risks of individual positions using complementary positions. Spreading portfolio capital across a variety of assets is known as diversification.

Risk is divided into two broad categories: idiosyncratic and systemic. Idiosyncratic risk is specific to an individual asset, sector, or position and can be minimized using diversification. For example, a portfolio containing only Apple stock is subject to risk factors specific to Apple and the tech sector. Some of those risks can be offset with the addition of an uncorrelated or inversely correlated asset, such as a commodity ETF like GLD. In this more diversified scenario, some hypothetical company-specific risk factors causing AAPL stock to depreciate may be reduced by the performance of GLD, which has relatively independent dynamics.

Comparatively, systemic risk is inherent to the market as a whole and cannot be diversified away. All traded assets are subject to systemic risk because every economy, market, sector, and company has the potential to fail. No amount of diversification will ever remove that element of uncertainty. Instead, the purpose of diversification is to construct a robust portfolio with minimal sensitivity to company-, sector-, or market-specific risk factors.

The process of building a diversified portfolio depends on the types of assets comprising the target portfolio. For an equity portfolio, the most effective way to diversify against idiosyncratic risk is to distribute portfolio capital across assets that have low or inversely correlated price movements. This is because the primary concern when trading equities is the directional movement of the underlying, specifically to the downside. Diversifying portfolio assets, typically using instruments for a variety of companies, sectors, and markets, reduces some of this directional concentration and improves the stability of the portfolio.

To understand the effectiveness of diversification by this method, consider the example outlined next. [Table 7.2](#) shows different portfolio allocation

percentages for two equity portfolios, [Table 7.3](#) shows the correlation of the assets in both portfolios, and [Figure 7.1](#) shows the comparative performance of the two portfolios. The historical directional tendencies are often estimated using the correlation coefficient, which quantifies the strength of the historical linear relationship between two variables. Recall that the correlation coefficient ranges from -1 to 1 , with 1 corresponding to perfect positive correlation, -1 corresponding to perfect inverse correlation, and 0 corresponding to no measured correlation.

Table 7.2 Two sample portfolios, each containing some percentage of market ETFs for reliable portfolio growth (SPY, QQQ), low volatility assets for diversification (GLD, TLT), and high volatility assets for increased profit potential (AMZN, AAPL).

	% Portfolio Allocation	
	Portfolio A	Portfolio B
Market ETFs	40%	50%
Low Volatility Assets	50%	0
High Volatility Assets	10%	50%

These portfolio weights were determined intuitively and not by any particular quantitative methodology. This example demonstrates the effectiveness of diversification rather than providing a specific framework for achieving diversification in equity portfolios.

Table 7.3 The five-year correlation history for the assets in Portfolios A and B. Though these relationships fluctuate with time over short timescales, they are assumed to remain relatively constant long term.

		Correlation (2015–2020)					
		SPY	QQQ	GLD	TLT	AMZN	AAPL
Market	SPY	1.0	0.89	−0.13	−0.33	0.62	0.64
ETFs	QQQ	0.89	1.0	−0.12	−0.26	0.75	0.74
Low							
Volatility	GLD	−0.13	−0.12	1.0	0.39	−0.12	−0.11
Assets	TLT	−0.33	−0.26	0.39	1.0	−0.18	−0.22
High							
Volatility	AMZN	0.62	0.75	−0.12	−0.18	1.0	0.50
Assets	AAPL	0.64	0.74	−0.11	−0.22	0.50	1.0

Table 7.2 outlines two portfolios: Portfolio A is a relatively diversified portfolio with conservative risk tolerances and moderate profit expectations, while Portfolio B is a risk tolerant and fairly concentrated portfolio. **Table 7.3** shows how the elements in Portfolio B (SPY, QQQ, AMZN, AAPL) have fairly high mutual historic correlations and therefore similar directional tendencies. Comparatively, half of Portfolio A is allocated to low volatility assets that are uncorrelated or inversely correlated with the market ETFs and high volatility assets. Therefore, due to the diversifying contributions of those relatively independent assets, Portfolio A is less sensitive to outlier market events. **Figure 7.1** shows how these portfolios would have performed from 2020–2021, importantly including the 2020 sell-off and subsequent recovery.

Portfolio Performance Comparison (2020–2021)

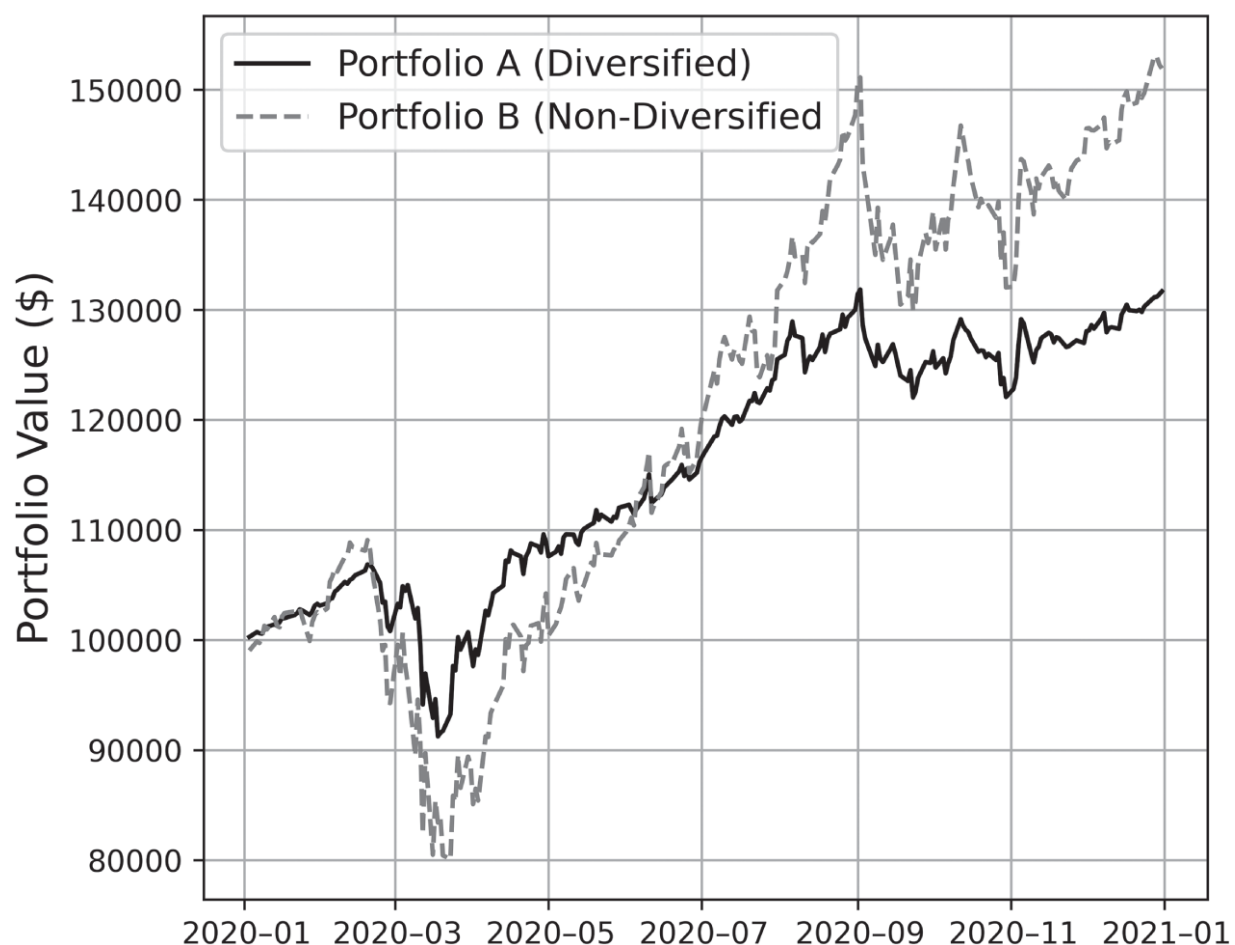


Figure 7.1 Performance comparison for Portfolios A and B from 2020 to 2021. Each portfolio begins with \$100,000 in initial capital.

Historic correlations have become *stronger* during financial crashes and sell-offs. Stated differently, assets have become more correlated or more inversely correlated during volatile market periods. The correlations in [Table 7.2](#), therefore, *underestimate* the correlation magnitudes that would have been measured from 2020–2021.

As a result of the COVID-19 pandemic, market ETFs and highly correlated assets, such as large cap tech stocks incurred significant drawdowns. Portfolio B, half of which was high volatile tech stocks, crashed by roughly 25% from February to late March 2020. Comparatively, Portfolio A still experienced massive drawdowns but only declined by 14% during the same period. Portfolio B is significantly more exposed to market volatility than Portfolio A, resulting in a more rapid, but unstable recovery following the

2020 sell-off. Throughout this year, Portfolio B grew by roughly 90% from its minimum in March while Portfolio A was growing by 44%, but Portfolio B was nearly twice as volatile. Nondiversified portfolios are generally more sensitive to sector- or market-specific fluctuations compared to diversified portfolios. Diversifying a portfolio across asset classes reduces position concentration risk and tends to reduce loss potential in the event of turbulent market conditions. However, [Figure 7.1](#) shows how more volatile, higher-risk portfolios can pay off with higher profits.

Due to their complex risk profiles, options are inherently more diversified relative to one another compared to their equity counterparts. Unlike equities, where the primary concern is directional risk, several factors may affect option P/L:

- Directional movement in the underlying price.
- Changes in IV.
- Changes in time to expiration.

Because exposure to each of these variables can be controlled according to the contract parameters, varying factors, such as duration/management time, underlying, and strategy creates an additional reduction in P/L correlation that is not possible when trading equities exclusively. However, diversifying against directional risk of the underlyings remains most essential from the perspective of risk management, particularly outlier risk management. Diversifying against nondirectional risk by varying strategy or contract duration is supplemental.

To understand why it is so essential to diversify the option underlyings of a portfolio, consider two market ETFs: SPY and QQQ. These assets have historically had highly correlated price dynamics and IV dynamics, as shown in the correlation matrix in [Table 7.4](#).

The equity underlyings and IV indices are highly correlated, meaning that IV expansion events and outlier price moves tend to happen simultaneously for these two assets. When such events do occur, short premium positions with these two underlyings may experience simultaneous tail losses. To get an idea of how often these positions have incurred simultaneous outlier losses historically, refer to the strangle statistics shown in [Table 7.5](#).

Table 7.4 Historic correlations between two market ETFs (SPY, QQQ) and the correlations between their implied volatility indices (VIX, VXN) from 2011 to 2020. Also included is the correlation between each market index and the respective VIX, for reference.

		Equity Price and IV Index Correlation (2011–2020)			
		SPY	QQQ	VIX	VXN
Equities	SPY	1.0	0.89	−0.80	
	QQQ	0.89	1.0		−0.76
Volatility	VIX	−0.80		1.0	0.89
Indices	VXN		−0.76	0.89	1.0

Table 7.5 The probability of outlier losses (worse than 200% of the initial credit) occurring simultaneously for 16 Δ SPY strangles and 16 Δ QQQ strangles from 2011 to 2020. All contracts have approximately the same duration (45 DTE), start date, and expiration date. The diagonal entries (SPY Strangle-SPY Strangle, QQQ Strangle-QQQ Strangle) indicate the probability of a strategy incurring an outlier loss individually, and the off-diagonal entries correspond to the probability of the pair incurring outlier losses simultaneously.

		Probability of Loss Worse than 200% (2011–2020)	
		SPY Strangle	QQQ Strangle
SPY Strangle		5.8%	3.9%
QQQ Strangle		3.9%	8.7%

Table 7.5 shows that it is reasonably unlikely for the pair of strategies to incur outlier losses simultaneously having occurred only 3.9% of the time. However, if these events were completely independent, then these compound losses would have occurred less than 1% of the time: $5.8\% \times 8.7\% \approx 0.50\%$. Additionally, when considering the outlier loss probability for each strategy on an individual basis, the effects of trading strangles with correlated underlyings becomes a bit clearer.

For example, the probability of a SPY strangle incurring an outlier loss is 5.8%. What is the probability a QQQ strangle will incur a simultaneous

outlier loss given that a SPY strangle has taken an outlier loss? To calculate this, one can use conditional probability.¹

$$P(\text{QQQ loss given SPY loss}) = \frac{P(\text{SPY loss and QQQ loss})}{P(\text{SPY loss})}$$

$$= \frac{3.9\%}{5.8\%} \approx 67\%$$

In other words, SPY strangles and QQQ strangles may only have simultaneous outlier losses 3.9% of the time, but when a SPY strangle incurs an outlier loss, there is a 67% *chance* that a QQQ strangle also will.² Generally, the probability of a compound loss is fairly low, but when one short premium position takes a loss there is often a high likelihood an equivalent position with a correlated underlying will experience a loss of comparable magnitude. Because the loss potential of these compound occurrences is so large, it is essential to diversify underlying equities and maintain appropriate position sizes for correlated options to reduce the likelihood and impact of compounding outlier losses.

Now consider two market ETFs (SPY and QQQ) and two diversifying ETFs that have been uncorrelated or inversely correlated to the market (GLD, TLT). The historic correlations are shown in [Table 7.6](#) and the probability of outlier losses occurring simultaneously are shown in [Table 7.7](#).

Table 7.6 Historic correlations among two market ETFs (SPY and QQQ), a gold ETF (GLD), and a bond ETF (TLT) from 2011 to 2020.

	Equity Price Correlation (2011–2020)			
	SPY	QQQ	GLD	TLT
SPY	1.0	0.89	−0.03	−0.41
QQQ	0.89	1.0	−0.04	−0.34
GLD	−0.03	−0.04	1.0	0.23
TLT	−0.41	−0.34	0.23	1.0

Table 7.7 The probability of outlier losses (worse than 200% of the initial credit) occurring simultaneously for different types of 16 Δ strangles held to expiration from 2011 to 2020. All contracts have approximately the same duration (45 DTE), open and close dates. The diagonal entries correspond to the probability of the specific strategy incurring an outlier loss individually, and the off-diagonal entries correspond to the probability of the pair incurring outlier losses simultaneously.

	Probability of Loss Worse than 200% for Different Strangles (2011–2020)			
	SPY	QQQ	GLD	TLT
SPY	5.8%	3.9%	2.1%	1.9%
QQQ	3.9%	8.7%	1.9%	1.7%
GLD	2.1%	1.9%	12%	4.8%
TLT	1.9%	1.7%	4.8%	12%

Again, it is relatively unlikely for any pair to incur simultaneous outlier losses, but this table shows the significant reduction in the *conditional* outlier probability when the underlying assets are uncorrelated or inversely correlated. Consider the following:

- If a SPY strangle incurs an outlier loss, there is a 67% chance of a compounding loss with a QQQ strangle.
- If a SPY strangle incurs an outlier loss, there is a 36% chance of a compounding loss with a GLD strangle.
- If a QQQ strangle incurs an outlier loss, there is a 20% chance of a compounding loss with a TLT strangle.

Compound losses still occur when the underlying assets have low or inversely correlated price movements, but this reduction in likelihood is crucial nonetheless. Having a portfolio that includes uncorrelated or inversely correlated assets is particularly meaningful during periods of unexpected market volatility when most assets develop a stronger correlation to the market and there are widespread expansions in IV. Though options can be diversified with respect to several variables, diversifying the underlying assets is one of the most effective ways to

reduce the impact of outlier events on a portfolio. Because diversification does not entirely remove the risk of compounding outlier losses, so maintaining small position sizes (at most 5% to 7% of portfolio capital allocated to a single position) remains critical.

Maintaining Portfolio Greeks

The Greeks form a set of risk measures that quantify different dimensions of exposure for options. Each contract has its own specific set of Greeks, but some Greeks have the convenient property of being additive across positions with different underlyings. Consequently, these metrics can be used to summarize the various sources of risk for a portfolio and guide adjustments. The following portfolio Greeks will be the focus of this section:

- Beta-weighted delta ($\beta\Delta$): Recall from [Chapter 1](#) that beta is a measure of systematic risk and specifically quantifies the directional tendency of the stock relative to that of the overall market. Stocks with positive correlation to the market have positive beta and stocks with negative correlation have negative beta. $\beta\Delta$ is similar to delta, which is the expected change in the option price given a \$1 change in the price of the underlying. When delta is beta-weighted, the adjusted value corresponds to the expected change in the option price given a \$1 change in some reference index, such as SPY.
- Theta (θ): The decline in an option's value due to the passage of time, all else being equal. This is generally represented as the expected decrease in an option's value per day.

Maintaining the balance of these two variables is crucial for the long-term health of a short options portfolio. $\beta\Delta$ represents the amount of directional exposure a position has relative to some index rather than the underlying itself. The cumulative portfolio $\beta\Delta$ delta represents the overall directional exposure of the portfolio relative to the market assuming that the beta index is a market ETF like SPY. Normalizing delta according to a standard underlying allows delta to be additive across all portfolio

positions. This *cannot* be done with unweighted delta because \$1 moves across different underlyings are not comparable, i.e., trying to add deltas of different positions is like adding inches and ounces. For example, a 50 Δ sensitivity to underlying A and a 25 Δ sensitivity to underlying B does not imply a 75 Δ sensitivity to anything, unless A and B happen to be perfectly correlated.

$\beta\Delta$ neutral portfolios are attractive to short premium traders because the portfolio is relatively insensitive to changes in the market, and profit is primarily driven by changes in IV and time. Adopting $\beta\Delta$ neutrality also simplifies aspects of the diversification process because a near-zero $\beta\Delta$ indicates low directional market exposure. As the delta of a position drifts throughout the contract duration, the overall delta of the portfolio is skewed. To maintain $\beta\Delta$ neutrality, existing positions can be re-centered (where the current trade is closed and reopened with a new delta), existing positions can be closed entirely, or new positions can be added. The most appropriate strategy depends on the current portfolio theta.

Theta is also additive across positions because the units of theta are identical for all options (\$/day). Because short premium traders consistently profit from time decay, the total theta across positions gives a reliable estimate for the expected daily growth of the portfolio. The theta ratio ($\text{theta/net portfolio liquidity}$) estimates the expected daily profit per unit of capital for a short premium portfolio. Options portfolios are subject to significant tail risk, so the expected daily profit should be significantly higher than a portfolio passively invested in the market to justify that risk. Therefore, one can determine the benchmark profit goals of an equivalent short options portfolio by referring to the daily P/L performance of a passively invested SPY portfolio as shown below in [Table 7.8](#).

Table 7.8 Daily performance statistics for five portfolios passively invested in SPY from 2011–2021. Each portfolio has \$100,000 in initial capital, and the amount of capital allocated in each portfolio ranges from 25% to 50%.

SPY Allocation Percentage	Daily Portfolio P/L (2011–2021)
25%	0.013%
30%	0.015%
35%	0.017%
40%	0.020%
50%	0.025%

From 2011–2021, a passively invested SPY portfolio collected between 0.013% and 0.025% daily depending on the percentage of capital allocated. In other words, these portfolios had daily profits between \$13 and \$25 per \$100,000 of capital over the past 10 years ($\$100,000 \times 0.00013$). However, the expected daily profit per unit of capital for a short options portfolio should be *significantly* higher than this benchmark. For most traders, the minimum theta ratio should range from 0.05% to 0.1% of portfolio net liquidity to justify the tail risks of short premium. In other words, short premium portfolios should have a daily expected profit between \$50 and \$100 per \$100,000 of portfolio buying power from **theta** decay.

The theta ratio should not exceed 0.2%. A higher theta ratio is preferable, but it should not be too high due to hidden gamma risk. Gamma (Γ) is the expected change in the option's delta given a \$1 change in the price of the underlying. Delta neutral positions are rarely gamma neutral, and if the gamma of a position is especially high, then the delta of the trade is highly sensitive to changes in the underlying price and is generally unstable. A position with high delta sensitivity can easily affect the overall $\beta\Delta$ neutrality of a portfolio.

The gammas of different derivatives cannot be compared across underlyings for similar reasons as to why raw delta cannot be compared across underlyings. Gamma cannot be accurately beta-weighted as delta can; however, a positive relationship between gamma and theta presents a solution to this problem. Positions with large amounts of theta, such as

trades with strikes that are close to at-the-money (ATM) or trades that are near expiration, typically also have large amounts of gamma risk. Because theta is additive across portfolio positions, the theta ratio is the most direct indicator of excessive gamma risk. This relationship between gamma and theta also demonstrates how short premium traders must balance the profitability of time decay with the P/L fluctuations resulting from gamma.

To summarize, the theta ratio for an options portfolio should range from 0.05% to 0.1% and should not exceed 0.2%. Based on the theta ratio and the amount of capital currently allocated, existing positions should then be re-centered, short premium positions should be added, or short premium positions should be removed. Given these benchmarks for expected daily profits, the procedure for modifying portfolio positions can be summarized as follows:

- If a properly allocated, a well-diversified portfolio is $\beta\Delta$ neutral but does not provide a sufficient amount of theta, then the positions in the portfolio should be reevaluated. In this case, perhaps some defined risk trades should be replaced with undefined risk trades or undefined risk positions should be rolled to higher deltas. New delta neutral positions can also be added, such as strangles and iron condors, for example. IV and theta are also highly correlated, meaning that higher IV underlyings could also be considered if theta is too low. These measures can be reversed if the portfolio has too much theta exposure while being $\beta\Delta$ neutral.
- If the theta ratio is too low (<0.1%), then either existing positions should be re-centered/tightened or new short premium positions should be added.

If the $\beta\Delta$ is too large and positive (bullish), then add new negative $\beta\Delta$ positions (e.g., add short calls on positive beta underlyings or add short puts on negative beta underlyings).

If the $\beta\Delta$ is too large and negative (bearish), then add new positive $\beta\Delta$ positions (e.g., add short puts on positive beta underlyings).

- If the theta ratio is too large ($>0.2\%$), then either existing positions should be re-centered/widened or short premium positions should be removed.

If the $\beta\Delta$ is too large and positive (bullish), then remove positive $\beta\Delta$ positions (e.g., remove short puts on positive beta underlyings).

If the $\beta\Delta$ is too large and negative (bearish), then remove negative $\beta\Delta$ positions (e.g., remove short calls on positive beta underlyings).

If a properly allocated, well-diversified portfolio provides a sufficient amount of theta but is not $\beta\Delta$ neutral, then existing positions should be reevaluated. For example, skewed positions could be closed and re-centered or replaced with new delta-neutral positions that offer comparable amounts of theta.

Takeaways

1. The amount of portfolio buying power allotted to short premium positions should range from 25% to 50% depending on the current market volatility, with the remaining capital either kept in cash or a low-risk passive investment. Of the amount allocated, at least 75% should be reserved for undefined risk trades (with no more than 7% allocated to a single position), and at most 25% should be reserved for defined risk strategies (with no more than 5% allocated to a single position). The total portfolio allocation guidelines *must* be followed, but there is more leniency for the per-trade allocation guidelines, especially in smaller accounts.
2. An options portfolio is typically composed of two types of positions: core and supplemental. Core positions are usually high-POP trades with moderate P/L variance that offer consistent profits and reasonable outlier exposure. Supplemental positions are not necessarily dependable sources of profit but rather tools for market engagement.

At most 25% of the capital allocated to short premium should go toward supplemental positions.

3. Unlike equity portfolios, options portfolios can be diversified with respect to multiple variables, such as duration/management time, underlying, and strategy. Diversifying the underlyings of an options portfolio remains the most essential diversification tool for portfolio risk management, particularly outlier risk management.
4. Beta-weighted delta ($\beta \Delta$) represents the amount of directional exposure a position has relative to some index rather than the underlying itself. Portfolio theta (θ) represents the expected daily growth of the portfolio. The minimum theta ratio for an options portfolio should range from 0.05% to 0.1% and should not exceed 0.2%. Maintaining the balance of these two Greeks ensures the risk-reward profile of an options portfolio remains as close to the target as possible.

Notes

- 1 For an introduction to conditional probability, refer to the appendix.
- 2 A 67% conditional probability of a compound loss is very high but lower than the compound loss probability when trading the equivalent equities. SPY and QQQ are *highly* correlated and experience near-identical drawdowns in periods of market turbulence. Therefore, that these options incur compound outlier losses only 70% of the time demonstrates the inherent diversification of options alluded to earlier.